Lecture 9

Vector fields and flows

Objectives. *A smooth vector field prescribes a velocity at every point of a manifold. Its integral curves describe the resulting motion. Comparing a second field along that motion gives the Lie bracket. We prove the connection between a zero bracket and commuting complete flows, then compute the rotation fields and their signs. Left translation turns a tangent vector at the identity of a Lie group into a vector field. This defines the group's Lie algebra and its exponential map.*

By the end of this lecture you will be able to:

- Find integral curves and identify their maximal time domains.
- Compute a Lie bracket and derive it from backward transport by a flow.
- Prove the commuting-flow criterion for complete fields.
- Reconcile rotation-field and angular-momentum commutator signs.

Unit 9A Fields, flows, brackets and rotations

9.1 Vector fields and related fields

Let M be a smooth n-manifold.

Definition 9.1 (Smooth vector field)**.** A *smooth vector field* on M is a smooth map

$$X\colon M \to TM, \qquad X(p) \in T_pM.$$

Such a map is called a *section* of the tangent bundle: composing it with the bundle projection gives the identity on M.

In local coordinates it has the form $X = X^i\partial_i$, where the coefficient functions X^i are smooth. Indeed, its bundle-chart expression is $x \mapsto (x, X^1(x), \dots, X^n(x))$, so these two smoothness conditions are equivalent.

The field acts on smooth real functions by

$$Xf = X^i\partial_i f.$$

This is a smooth function. The product rule gives $X(fg) = (Xf)g + f(Xg)$, so X acts as a derivation of functions. The value $(Xf)(p)$ is the point derivation $X_p(f)$ of Lecture 8. The same definition applies to smooth functions on any open subset of M.

Definition 9.2 (Push-forward field)**.** For a diffeomorphism $f\colon M \to N$, the *push-forward*

field f_*X on N is given by

$$(f_*X)_q = \mathrm{d}f_{f^{-1}(q)}X_{f^{-1}(q)}.$$

The inverse selects exactly one starting point for each q.

Definition 9.3 (Related fields). For a general smooth map f, fields X on M and Y on N are *f-related* when $\mathrm{d}f_pX_p = Y_{f(p)}$ for every $p \in M$.

We use related fields to compare motion on different manifolds. For the inclusion $i\colon S^1 \hookrightarrow \mathbb{R}^2$, write $i(\theta) = (\cos\theta, \sin\theta)$ in an angular chart. Its differential gives

$$\mathrm{d}i_\theta(\partial_\theta) = (-\sin\theta, \cos\theta).$$

Thus ∂_θ is i-related to $x\partial_y - y\partial_x$.

A field need not descend under a general map. For $f\colon \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ and $X = \partial_x$, the image vector is $2x\partial_y$ at $y = x^2$. The starting points 1 and -1 both map to 1 but give opposite nonzero vectors there. Thus no f-related field Y exists on the target. Lecture 10 compares this restriction with the pullback of forms.

9.2 Integral curves and maximal flows

Definition 9.4 (Integral curve). An *integral curve* of X is a smooth curve $\gamma\colon I \to M$ on an open time interval satisfying $\gamma'(t) = X_{\gamma(t)}$. Here $\gamma'(t)$ is the tangent class at $\gamma(t)$ of $s \mapsto \gamma(t+s)$, as in Lecture 8. For initial point p, impose $0 \in I$ and $\gamma(0) = p$. In coordinates this is the ordinary differential equation (ODE)

$$\dot{x}^i(t) = X^i(x(t)).$$

Definition 9.5 (Maximal curve and maximal flow). A curve is *maximal* if it has no extension to a larger open time interval as a solution with that initial condition. The resulting solution map is the *maximal flow*, written $\varphi\colon \mathcal{D} \to M$, with $\varphi_t(p) = \gamma_p(t)$. Its time domain at p is the interval $I_p = \{t : (t,p) \in \mathcal{D}\}$.

Definition 9.6 (Complete field). A field is *complete* if $I_p = \mathbb{R}$ for every $p \in M$.

Definition 9.7 (Support). The *support* of a field is $\operatorname{supp} X = \overline{\{p : X_p \neq 0\}}$. The field has *compact support* when this closed set is compact.

For translation and dilation on $\mathbb{R}$, direct differentiation gives

$$X = \partial_x: \quad \varphi_t(x) = x + t, \qquad X = x\partial_x: \quad \varphi_t(x) = e^t x.$$

Both flows are defined for every real time. Their time maps satisfy the addition law by addition of real numbers and multiplication of exponentials.

The planar dilation field $x\partial_x + y\partial_y$ has flow $\varphi_t(x,y) = e^t(x,y)$. The figure shows its velocities and one integral curve.

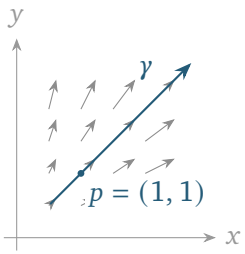


Figure 9.1: The field (x, y) and its integral curve $e^t(1, 1)$. Each arrow gives a velocity at its own point; the marked curve follows those velocities.

On $\mathbb{R}$, the field $X = x^2\partial_x$ has maximal flow

$$\varphi_t(x) = \frac{x}{1 - tx}, \qquad \mathcal{D} = \{(t, x) : 1 - tx > 0\}.$$

Differentiating gives $\partial_t \varphi_t = \varphi_t^2$, and the initial value is x. At $x = 0$ the solution is constant for all time. For $x > 0$ it diverges at $t = 1/x$. For $x < 0$ the divergence is at that negative time. These intervals are maximal because a continuous real-valued extension cannot pass through a divergence. Requiring only a nonzero denominator would include a separate branch beyond the pole. This field has noncompact support. A linear field $X_A(x) = Ax$ on $\mathbb{R}^n$, where $A \in \mathbb{R}^{n\times n}$ is nonzero, also has noncompact support. By contrast, its flow $x \mapsto e^{tA}x$ (§9.3) is defined for all real t. The graph shows the failure at $x = 1$ for $x^2\partial_x$.

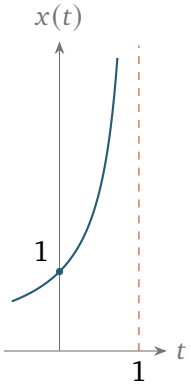


Figure 9.2: The solution starting at $x = 1$ diverges as t approaches 1 from below. The rational expression beyond the pole is not part of this integral curve.

Theorem 9.8 (Flow properties · *cited; laws and completeness proved from the cited flow properties*)**.** *A smooth vector field has a unique smooth maximal flow on an open set $\mathcal{D} \subseteq \mathbb{R} \times M$ containing $\{0\} \times M$. The time domains I_p are intervals. Its time maps satisfy*

$$\varphi_0 = \mathrm{id}, \qquad \varphi_{s+t}(p) = \varphi_s(\varphi_t(p))$$

whenever both sides are defined. Each time map is a diffeomorphism between its open source and image, with inverse φ_{-t}. Every smooth field with compact support is complete.

Existence and uniqueness of maximal curves, openness of $\mathcal{D}$ and joint smoothness in (t, p) are cited from Lee, second edition, Theorem 9.12. Its local input is the ODE theorem in charts. We prove the laws and completeness from these properties.

Proof. First fix $t \in I_p$ and put $q = \varphi_t(p)$. The curve $u \mapsto \varphi_{t+u}(p)$ is an integral curve starting at q. Uniqueness identifies it with $u \mapsto \varphi_u(q)$ on their common interval. Maximality gives $I_q = I_p - t$: either larger interval would, by shifting back and using uniqueness, extend the other solution. This proves the group law and ensures that $\varphi_{-t}(q) = p$. Define $M_t = \{p \in M : t \in I_p\}$, open because $\mathcal{D}$ is open. Applying the inverse identity with $-t$ shows that $\varphi_t \colon M_t \to M_{-t}$ is onto and both inverse compositions are identities. Joint smoothness gives smooth time maps and inverses.

Next we use compactness to obtain a common positive existence time. This permits continuation from every point reached by a solution. Let $K = \operatorname{supp} X$ be compact. If K is empty, the field is zero and every solution is constant. Otherwise, openness of $\mathcal{D}$ gives, near each $p \in K$, an open set U_p and a time $\varepsilon_p > 0$ for which the flow is defined on $(-\varepsilon_p, \varepsilon_p) \times U_p$. Choose a finite subcover of K and let $\delta > 0$ be the minimum of these finitely many times. Solutions starting in K exist for at least time δ in both directions. Outside K, the constant curve is a solution and uniqueness makes it the maximal solution. Thus every starting point has a solution for $|t| < \delta$. If a maximal interval had a finite right endpoint b, restart the solution at a time $t < b$ with $b - t < \delta$. The new solution extends past b, contradicting maximality and uniqueness. The same argument at a finite left endpoint proves completeness. ∎

In particular, every smooth field on a compact manifold is complete, since its support is a closed subset of that manifold.

On the torus $T^2 = S^1 \times S^1$, local angular coordinates (θ^1, θ^2) change by constant multiples of 2π. Their coordinate fields therefore define global fields. Fix $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and set

$$X = \partial_{\theta^1} + \alpha \partial_{\theta^2}, \qquad \varphi_t(e^{\mathrm{i}\theta^1}, e^{\mathrm{i}\theta^2}) = (e^{\mathrm{i}(\theta^1 + t)}, e^{\mathrm{i}(\theta^2 + \alpha t)}).$$

Differentiating the angular coordinates gives the prescribed velocity. Starting at $(1, 1)$ gives $\gamma(t) = (e^{\mathrm{i}t}, e^{\mathrm{i}\alpha t})$, the irrational curve of Lecture 7. The formula exists for every $t \in \mathbb{R}$. Completeness concerns the time domain, even though this orbit is an injective immersion whose image is not embedded.

9.3 The Lie bracket and matrix signs

We compare two motions by following each and then reversing each in the same order. On $\mathbb{R}^2$, take $X = \partial_x$ and $Y = x\partial_y$. Their flows are $\varphi_u(x, y) = (x + u, y)$ and $\psi_v(x, y) = (x, y + vx)$. Starting at (x, y), the successive endpoints for times $u, v, -u, -v$ are

$$\begin{aligned}(x, y) &\longmapsto (x + u, y) \longmapsto (x + u, y + (x + u)v) \\ &\longmapsto (x, y + (x + u)v) \longmapsto (x, y + uv).\end{aligned}$$

The final gap is exactly $uv\partial_y$. The figure compares this path with commuting coordinate translations. The shear panel uses $x = u = v = 1$ and $y = 0$.

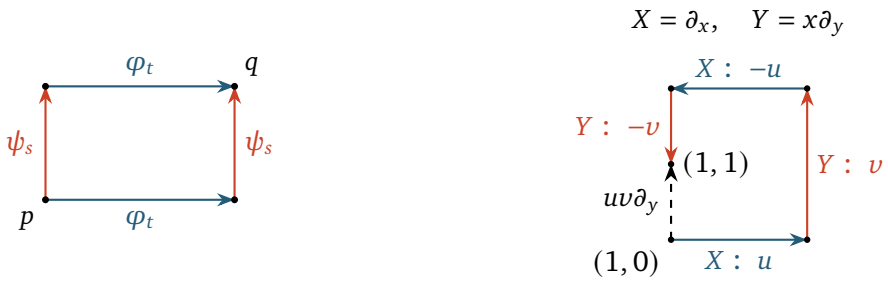


Figure 9.3: Coordinate translations close the rectangle. For $X = \partial_x$ and $Y = x\partial_y$, the return segment is shorter and leaves a positive vertical gap.

Definition 9.9 (Lie bracket). For fields X and Y, define their *Lie bracket* by its action on smooth functions:

$$[X, Y]f = X(Yf) - Y(Xf).$$

The first composition expands as

$$X(Yf) = X^i(\partial_i Y^j)\partial_j f + X^i Y^j \partial_i \partial_j f.$$

In the second composition, swap the indices i and j in its second-derivative term. Equality of mixed partial derivatives makes those terms cancel. The result is

$$[X, Y] = \big(X^i \partial_i Y^j - Y^i \partial_i X^j\big)\partial_j.$$

An individual composition XY can contain second derivatives of f. Their cancellation makes the commutator a first-order derivation. The coefficients are smooth, so this expression is a smooth vector field. Its action on functions is intrinsic, making it independent of coordinates. Bilinearity, antisymmetry and Jacobi follow from the commutator calculation for operators, exactly as in Lecture 8.

For general smooth fields, let $f_{u,v}$ be the map obtained by following X, Y, X, Y for times $u, v, -u, -v$, where these local flows are defined. In a fixed chart near p, the displacement satisfies

$$f_{u,v}(p) - p = uv[X, Y]_p + O\big(|uv|(|u| + |v|)\big) \qquad (u, v \to 0).$$

Indeed, differentiating the four compositions once in each time at zero gives $DY_p X_p - DX_p Y_p$. The displacement vanishes when either time is zero. Smoothness therefore factors it as uv times a smooth function of the times. That function has value $[X, Y]_p$ at zero, which gives the stated remainder. For the shear fields above, the bracket is ∂_y and the remainder vanishes.

The bracket preserves related fields under any smooth map f. Suppose X_j is f-related to Y_j for $j = 1, 2$. For every local smooth function h on the target, the chain rule gives $X_j(h \circ f) = (Y_j h) \circ f$. Applying it twice yields

$$[X_1, X_2](h \circ f) = ([Y_1, Y_2]h) \circ f.$$

Testing coordinate functions proves that the brackets are f-related. For a diffeomorphism this says $[f_* X, f_* Y] = f_*[X, Y]$.

Proposition 9.10 (Linear fields and the matrix commutator). *For $A, B \in \mathbb{R}^{n\times n}$, the fields $X_A(x) = Ax$ on $\mathbb{R}^n$ satisfy*

$$[X_A, X_B] = X_{-[A,B]}, \qquad [A, B] = AB - BA.$$

Proof. For linear fields, $DX_A = A$ and $DX_B = B$. The coordinate formula gives $[X_A, X_B](x) = BAx - ABx$, proving the minus sign. ∎

For example, let $A = E_{12}$ and $B = E_{21}$ in two dimensions, where E_{ij} has entry 1 in position (i, j) and zero elsewhere. Their fields and brackets are

$$X_A = y\partial_x, \qquad X_B = x\partial_y, \qquad [X_A, X_B] = -x\partial_x + y\partial_y.$$

The matrix commutator is $[A, B] = \operatorname{diag}(1, -1)$. Its linear field is the negative of the displayed bracket.

Recall the matrix exponential from *Linear Algebra*. The solution of $\dot{x} = Ax$ with initial value x is $e^{tA}x$. Thus the linear-field flow multiplies the position column on the left. For a smooth function f, the iterated field action is

$$(X_A X_B f)(x) = \left.\frac{\partial^2}{\partial s\,\partial t}\right|_{s=t=0} f(e^{tB}e^{sA}x).$$

The field X_B acts first on f, but the matrix e^{sA} acts first on x. This reversal of composition order explains the minus sign.

9.4 The Lie derivative of a field

Vectors at different points cannot be subtracted intrinsically. For $t \in I_p$, the map $\varphi_{-t}\colon M_{-t} \to M_t$ is a diffeomorphism and $\varphi_t(p) \in M_{-t}$. Thus it pushes $Y_{\varphi_t(p)}$ back to p by $\mathrm{d}\varphi_{-t}|_{\varphi_t(p)}$.

Definition 9.11 (Lie derivative). The *Lie derivative* is the resulting derivative in T_pM:

$$(L_X Y)_p = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 ((\varphi_{-t})_* Y)_p.$$

For functions, define $L_X f = Xf$, the derivative of $f(\varphi_t(p))$ at zero.

We compare the two vectors only after both have base point p. The figure shows this comparison for the shear fields at $p = (1, 0)$ and $t = 1$.

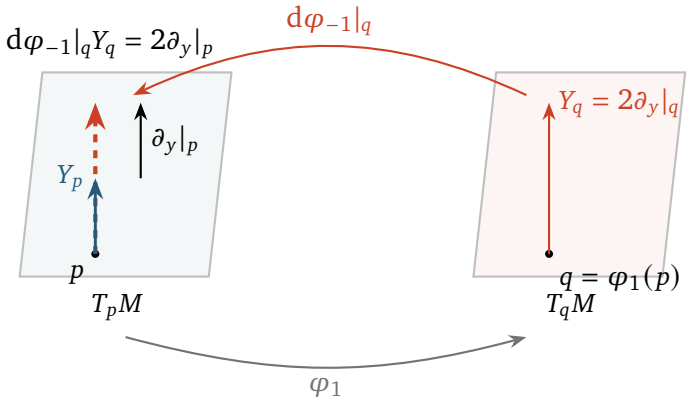


Figure 9.4: The vector Y_q is based at $q = \varphi_t(p)$. Its backward image and Y_p are both based at p; their difference divided by t tends to $L_X Y$.

For $X = \partial_x$ and $Y = x\partial_y$, the flow is $\varphi_t(x, y) = (x + t, y)$. The inverse flow has identity Jacobian, so

$$\mathrm{d}\varphi_{-t}|_{\varphi_t(p)} Y_{\varphi_t(p)} = (x + t)\partial_y|_p, \qquad \frac{(x + t)\partial_y|_p - x\partial_y|_p}{t} = \partial_y|_p \quad (t \neq 0).$$

The derivative is $L_X Y = \partial_y$, in agreement with the bracket.

Theorem 9.12 (Flow derivative and bracket · *proved given local flow existence and smooth dependence*)**.** *The bracket $[X, Y]$ is a smooth vector field. The Lie derivative exists and satisfies $L_X Y = [X, Y]$.*

Proof. The bracket's coordinate formula above proves its smoothness. The change in Y along the trajectory contributes $DY_p X_p$. The changing inverse differential contributes $-DX_p Y_p$. In a coordinate neighbourhood, the ODE gives $\varphi_t(p) = p + tX(p) + O(t^2)$. Differentiating the ODE $\partial_t \varphi_t = X(\varphi_t)$ in p, as joint smoothness permits, gives $\partial_t D\varphi_t = DX(\varphi_t) D\varphi_t$. Hence $D\varphi_t(p) = I + tDX_p + O(t^2)$. Inverting gives the inverse Jacobian $I - tDX_p + O(t^2)$ at $\varphi_t(p)$. Thus

$$\begin{aligned}((\varphi_{-t})_* Y)_p &= \big(I - tDX_p + O(t^2)\big)\big(Y_p + tDY_p X_p + O(t^2)\big) \\ &= Y_p + t\big(DY_p X_p - DX_p Y_p\big) + O(t^2).\end{aligned}$$

The coefficient of t is the bracket's coordinate formula. The definition by intrinsic push-forward makes this computation independent of the chosen neighbourhood. ∎

This is Tu, Theorem 20.4, with the same backward-push convention. Forward push would give the opposite sign.

9.5 Commuting complete flows

Suppose X and Y have flows φ_t and ψ_s.

Definition 9.13 (Commuting fields and flows)**.** The fields *commute* when $[X, Y] = 0$. Their complete flows commute when $\varphi_t \circ \psi_s = \psi_s \circ \varphi_t$ for every $s, t \in \mathbb{R}$.

Theorem 9.14 (Commuting flows · *proved given flow existence, uniqueness and smooth dependence*)**.** *For two complete smooth fields, their bracket vanishes if and only if their flows commute for all real times.*

Proof. The flow group law, the push-forward chain rule $(f \circ g)_* = f_* g_*$ and Theorem 9.12 give

$$\frac{\mathrm{d}}{\mathrm{d}t}(\varphi_{-t})_* Y = (\varphi_{-t})_*[X, Y].$$

Indeed, write the value at $t + h$ as $(\varphi_{-t})_*((\varphi_{-h})_* Y)$ and differentiate in h at zero. If the bracket is zero, this family is constant and equals Y. Hence $(\varphi_t)_* Y = Y$ for every t. For fixed t, both curves $s \mapsto \varphi_t(\psi_s(p))$ and $s \mapsto \psi_s(\varphi_t(p))$ solve the Y equation with the same initial point. Uniqueness makes them equal. Completeness gives the equality for all $s, t \in \mathbb{R}$.

Conversely, differentiate commuting flows in s at zero. The result is $\mathrm{d}\varphi_t|_p Y_p = Y_{\varphi_t(p)}$, so $(\varphi_{-t})_* Y = Y$. Differentiating in t at zero gives $[X, Y] = 0$. ∎

9.6 Distributions and Frobenius's theorem *(optional)*

Later material on rotations, differential forms and connections does not depend on this section. We test whether prescribed tangent planes are tangent to local surfaces. Frobenius gives a criterion for their local existence.

Definition 9.15 (Distribution)**.** A *smooth rank-k distribution* assigns a k-dimensional subspace $D_p \subseteq T_pM$ to every point, locally spanned by k independent smooth fields.

Definition 9.16 (Involutive distribution)**.** A distribution is *involutive* when the bracket of any two local fields taking values in D also takes values in D.

Definition 9.17 (Integral manifold)**.** An *integral manifold* is an immersion $i\colon S \to M$ satisfying

$$\mathrm{d}i_s(T_sS) = D_{i(s)} \qquad (s \in S).$$

Theorem 9.18 (Frobenius · *cited*)**.** *Every smooth involutive distribution of constant rank k has, near each point, coordinates in which it is spanned by $\partial_1, \dots, \partial_k$. Fixing the other coordinates gives local integral manifolds.*

This is Lee, second edition, Theorem 19.12. For example, the planes spanned by ∂_x and ∂_y in $\mathbb{R}^3$ have integral surfaces $z = c$. By contrast, consider the rank-two distribution on $\mathbb{R}^3$ spanned by

$$X = \partial_x + y\partial_z, \qquad Y = \partial_y, \qquad [X, Y] = -\partial_z.$$

The fields are independent at every point, as their x and y components show. Matching those components in $aX + bY = -\partial_z$ would force $a = b = 0$. Thus the bracket lies outside the prescribed plane at every point.

No local immersed surface can have these tangent planes. Restricted to a small enough open set, an immersion is an embedding: by the inverse-function argument of Lecture 7, d of its target coordinates are local coordinates on its d-dimensional source. On such a surface the two fields would restrict to fields related by inclusion to X and Y. Naturality would make their bracket tangent too, a contradiction. Equivalently, projection to (x, y) is locally invertible on a candidate surface. Writing that surface as $z = h(x, y)$ would require $h_x = y$ and $h_y = 0$. The two mixed derivatives would be 1 and 0, contradicting smoothness. The figure illustrates the same obstruction by exact flows in the prescribed planes. The drawing uses $p = (0, 0, 0)$ and $u = v = 1$.

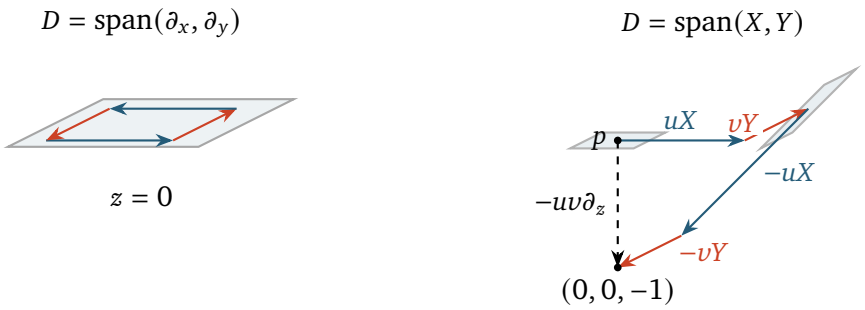


Figure 9.5: Horizontal planes are tangent to $z = c$. For $\mathrm{span}(\partial_x + y\partial_z, \partial_y)$, the four-flow path ends at $(x, y, z - uv)$; the vertical gap illustrates the computed bracket obstruction.

The paths in the second panel have successive endpoints

$$(x,y,z) \longmapsto (x+u,y,z+uy) \longmapsto (x+u,y+v,z+uy)$$
$$\longmapsto (x,y+v,z-uv) \longmapsto (x,y,z-uv).$$

The irrational field on T^2 spans a rank-one distribution. Its local sections have the form fX, and the product rule gives $[fX,gX] = (fXg - gXf)X$, so the distribution is involutive. Its orbits are integral manifolds. The irrational orbit of Lecture 7 is an injective immersion but is not embedded. This example explains why the definition requires only an immersion.

9.7 Rotation fields and angular momentum

In Cartesian coordinates on $\mathbb{R}^3$, the rotation fields are

$$L_x = y\partial_z - z\partial_y, \qquad L_y = z\partial_x - x\partial_z, \qquad L_z = x\partial_y - y\partial_x.$$

For example, $L_x(x^2+y^2+z^2) = 2yz - 2zy = 0$. Cyclic permutation gives the same result for the other fields. Thus each is tangent to every sphere centred at zero. Its integral curves retain that radius. Here L_x names a rotation field. The subscript X in the Lie derivative L_X names the generating vector field. In cylindrical coordinates $x = r\cos\phi$ and $y = r\sin\phi$ with $r > 0$, differentiating at fixed r and z gives $\partial_\phi = -y\partial_x + x\partial_y = L_z$. The angle ϕ is the azimuth. Thus the L_z flow adds t to the azimuth. On S^2, this field vanishes exactly at the two poles. The arrows in the figure show the azimuthal velocities.

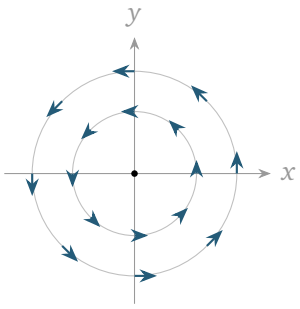


Figure 9.6: The field $L_z = (-y, x, 0)$ has circular trajectories in each horizontal plane. The arrows are its tangent velocities; the axis consists of fixed points.

Example 9.19 (Rotation flows and brackets)**.** The three flows are defined for all real t:

$$\varphi_t^x(x,y,z) = (x,\ y\cos t - z\sin t,\ y\sin t + z\cos t),$$
$$\varphi_t^y(x,y,z) = (x\cos t + z\sin t,\ y,\ -x\sin t + z\cos t),$$
$$\varphi_t^z(x,y,z) = (x\cos t - y\sin t,\ x\sin t + y\cos t,\ z).$$

For the first flow, differentiating its last two components gives $(-y\sin t - z\cos t,\ y\cos t - z\sin t)$, which is $(-z(t), y(t))$. At $t = 0$ the components are the initial coordinates. The same differentiation in the other coordinate planes verifies the other flows. Thus all three restrict to complete flows on S^2.

Their linear matrices are the basis matrices $A(1,0,0)$, $A(0,1,0)$ and $A(0,0,1)$ of Lecture 8. For a unit axis $n \in \mathbb{R}^3$, Lecture 8's matrix is $A(n)w = n \times w$. Lecture 2's axis–angle rotation $R(tn)$ has initial value I and derivative $A(n)R(tn)$. Uniqueness of the linear ODE

therefore gives $e^{tA(n)} = R(tn)$. The displayed flows are these matrix exponentials for the three coordinate axes.

One bracket can also be read directly on coordinates: $[L_x, L_y]x = L_x(z) = y$, $[L_x, L_y]y = -L_y(-z) = -x$ and $[L_x, L_y]z = 0$. Therefore

$$[L_x, L_y] = -L_z, \qquad [L_y, L_z] = -L_x, \qquad [L_z, L_x] = -L_y.$$

The matrices have the positive cyclic commutator, so this is precisely the linear-field minus sign of Proposition 9.10.

Physical input. For a scalar particle wavefunction, orbital angular momentum about axis a is represented by the differential operator $\widehat{L_a} = -\mathrm{i}\hbar L_a$ for $a \in \{x, y, z\}$. The constant $\hbar$ is the reduced Planck constant.

On complex smooth functions, these angular-momentum differential operators have the commutator

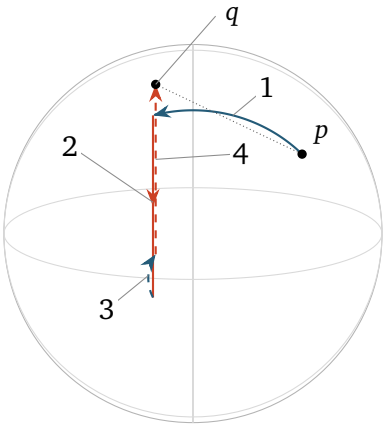
$$[\widehat{L_x}, \widehat{L_y}] = (-\mathrm{i}\hbar)^2(-L_z) = \hbar^2 L_z = \mathrm{i}\hbar\widehat{L_z}.$$

Thus the rotation-field sign agrees with the positive quantum angular-momentum relation. This is the differential-operator identification; no quantum representation theorem is assumed.

The rotation flows about different axes do not commute. For instance, the two orders of quarter-turns give

$$\varphi^x_{\pi/2}\big(\varphi^y_{\pi/2}(1,0,0)\big) = (0,1,0), \qquad \varphi^y_{\pi/2}\big(\varphi^x_{\pi/2}(1,0,0)\big) = (0,0,-1).$$

For small times, following L_x, L_y, L_x, L_y for $u, v, -u, -v$ has displacement $-uvL_z + O(|uv|(|u| + |v|))$ in a local chart. At $(1,0,0)$ its leading term is $(0, -uv, 0)$. The figure shows the corresponding four arcs at finite times on the unit sphere.



$1: L_x, \quad 2: L_y, \quad 3: -L_x, \quad 4: -L_y$

Figure 9.7: Successive rotations about x, y, x, y for times $\pi/3, \pi/3, -\pi/3, -\pi/3$ start at $p = (1,1,1)/\sqrt{3}$ and end at $q \neq p$. Solid arcs show the first two rotations and dashed arcs their reversals. Arcs 2 and 4 lie on nearly the same circle and are told apart by dashing. The dotted chord joins the final point to the initial point; all arcs are exact rotation trajectories.

Self-check. Find the flow of $x\partial_y - y\partial_x$ on $\mathbb{R}^2$ and its value at time $\pi/2$ from $(1,0)$. Include the trajectory from the origin.

Answer. The solution is $(x\cos t - y\sin t, x\sin t + y\cos t)$ for every $t \in \mathbb{R}$. From $(1,0)$ it reaches $(0,1)$ at time $\pi/2$, so positive time gives counter-clockwise rotation. Differentiating the displayed solution gives $(-y(t), x(t))$, which verifies the prescribed field. The origin is a constant solution.

Self-check. For $X = x\partial_x + 2y\partial_y$ and $Y = x^2\partial_x + xy\partial_y$ on $\mathbb{R}^2$, compute $L_X Y$ from backward transport. Verify the result by the coordinate bracket.

Answer. The flow is $\varphi_t(x,y) = (e^t x, e^{2t}y)$. Its inverse differential is $\operatorname{diag}(e^{-t}, e^{-2t})$. Consequently

$$((\varphi_{-t})_* Y)_{(x,y)} = \begin{pmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{pmatrix} \begin{pmatrix} e^{2t}x^2 \\ e^{3t}xy \end{pmatrix} = e^t \begin{pmatrix} x^2 \\ xy \end{pmatrix}.$$

Differentiating at zero gives $L_X Y = x^2\partial_x + xy\partial_y$. The bracket has x component $2x^2 - x^2 = x^2$ and y component $3xy - 2xy = xy$, so it agrees. The inverse differential supplies the subtractions in both components.

Unit 9B Left-invariant fields and the exponential

9.8 Left-invariant fields and their bracket

Lecture 8 put the matrix commutator on the tangent space at the identity of a matrix group. On an abstract Lie group, we extend a tangent vector at the identity to a field by left translation. Its field bracket then defines a bracket on that tangent space. For $\mathrm{SO}(3)$ this constructs fields on the group itself, distinct from the rotation fields on $\mathbb{R}^3$.

For a Lie group, write $\ell_g(h) = gh$ for left translation.

Definition 9.20 (Left-invariant field)**.** A field is *left-invariant* when $(\ell_g)_* X = X$ for all $g \in G$.

Here a *matrix Lie group* means an embedded matrix subgroup with the smooth structure of Lecture 8. For tangent matrices, the formula $A^L(g) = gA$ follows by differentiating left multiplication.

Proposition 9.21 (Left-invariant matrix fields)**.** *For tangent matrices A and B at the identity of a matrix Lie group, the left-invariant fields $A^L(g) = gA$ satisfy $[A^L, B^L] = [A,B]^L$.*

Proof. The group fields are related by the inclusion i to the ambient fields $g \mapsto gA$ and $g \mapsto gB$. Their ambient derivatives in direction K are KA and KB. The related-fields result identifies their ambient bracket with the inclusion of the group bracket. Thus

$$\mathrm{d}i_g([A^L, B^L]_g) = (gA)B - (gB)A = g[A,B].$$

The inclusion has injective differential, so this identifies the group bracket. ∎

The map $x \mapsto Ax$ multiplies a column on the left. The map $g \mapsto gA$ multiplies a matrix on the right. The coordinate bracket therefore gives $BA - AB$ for the first map and $g(AB - BA)$ for the second. In particular, $A(1,0,0)$ and $A(0,1,0)$ have positive cyclic matrix commutator, their left-invariant fields have positive bracket, and their rotation fields have negative bracket.

For an abstract Lie group, each $A \in T_eG$ determines a left-invariant field by $A^L(g) = \mathrm{d}(\ell_g)_e A$. Smoothness follows by differentiating the smooth multiplication map in its second argument. Associativity and the chain rule give

$$(\ell_g)_* A^L(h) = \mathrm{d}(\ell_g)_h \mathrm{d}(\ell_h)_e A = \mathrm{d}(\ell_{gh})_e A = A^L(gh).$$

This proves invariance, and evaluation at the identity recovers A. Conversely invariance forces this formula, so the correspondence is a linear bijection. Naturality of the bracket makes $[A^L, B^L]$ left-invariant.

Definition 9.22 (Lie algebra of a Lie group)**.** The Lie algebra of G is T_eG with bracket

$$[A, B] = [A^L, B^L](e).$$

The field bracket's bilinearity, antisymmetry and Jacobi identity give these properties to the bracket at the identity. For matrix groups it agrees with the commutator just computed.

For example, the group of maps $r \mapsto ar+b$, where $a > 0$, has law $(a, b)(a', b') = (aa', b+ab')$. Set $H = \partial_a$ and $N = \partial_b$ at the identity $(1, 0)$. Differentiating left translation gives $H^L = a\partial_a$ and $N^L = a\partial_b$. The coordinate bracket gives $[H^L, N^L] = a\partial_b = N^L$.

9.9 One-parameter subgroups and the exponential

Definition 9.23 (One-parameter subgroup)**.** A *one-parameter subgroup* is a smooth map $c\colon \mathbb{R} \to G$ with $c(s+t) = c(s)c(t)$ and $c(0) = e$.

The integral curve of A^L starting at the identity is a one-parameter subgroup. To prove this, take a local solution on $(-\delta, \delta)$ starting at the identity. Left translation gives a solution on that same interval starting at every group element. The uniform-time continuation argument in the proof of Theorem 9.8 proves completeness. Left invariance and uniqueness identify the two curves $t \mapsto c(s+t)$ and $t \mapsto c(s)c(t)$, proving the group law. Conversely, differentiating a one-parameter subgroup's law makes it an integral curve of A^L, with $A = c'(0)$. For matrices the solution is $c(t) = e^{tA}$, where

$$e^{tA} = \sum_{k=0}^{\infty} \frac{t^k A^k}{k!}.$$

If each $|A_{ij}| \le m$, multiplication of matrices gives $|(A^k)_{ij}| \le n^{k-1} m^k$ for $k \ge 1$, by induction. The resulting scalar exponential bounds the series and its differentiated series uniformly on bounded time intervals. Termwise differentiation gives $c'(t) = c(t)A$ and $c(0) = I$. The group-valued curve, viewed through the inclusion, solves this ambient equation. Ambient uniqueness identifies it with the series. The full field flow is $g \mapsto ge^{tA}$. For this right-multiplication flow, iterated differentiation gives

$$(A^L B^L f)(g) = \left.\frac{\partial^2}{\partial s\, \partial t}\right|_{s=t=0} f(g e^{sA} e^{tB}).$$

The order of the field actions agrees with the order of these right factors. This explains the positive sign of the left-invariant bracket.

Definition 9.24 (Lie group exponential)**.** For $A \in T_eG$, let c_A be the integral curve of A^L with $c_A(0) = e$. The *exponential* is the map

$$\exp\colon T_eG \longrightarrow G, \qquad \exp A = c_A(1).$$

Rescaling time in the integral-curve equation gives $c_A(t) = \exp(tA)$. For a matrix group, the preceding series calculation identifies this map with the matrix exponential.

Self-check. State in one sentence what the Lie algebra of an abstract Lie group is and why left-invariant fields enter its definition.

Answer. The Lie algebra is T_eG with the bracket $[A, B] = [A^L, B^L](e)$, obtained by extending tangent vectors by left translation so that their vector-field bracket is defined. Naturality makes that bracket left-invariant, so evaluation at the identity returns a tangent vector in the same space.

Summary. A vector field prescribes an ordinary differential equation in every chart. Uniqueness gives the flow group law, and compact support gives completeness. The Lie derivative of a field uses backward push-forward and equals its bracket with the generating field. Complete fields have zero bracket exactly when their flows commute. Linear fields Ax have the negative matrix commutator, whereas left-invariant matrix fields gA have the positive one. Lecture 10 introduces forms and their pullback under arbitrary smooth maps.

Reading. Tu, *An Introduction to Manifolds*, §14 and §20.2; Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 9.12 and 19.12.